



HOW CONSTRUCTIONS MEAN

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What are constructions?

- Traditionally: recipes for word combination used to meet specific communicative goals (issuing orders, requesting information, attributing a property to an entity).
- In Construction Grammar (CxG), constructions are descriptions of **constructs** (local trees).
- A **listeme** is a 'listed' description of a word or idiomatically combining form.
- Listemes and constructions describe classes of linguistic model objects (signs, constructs).
- There are no transformations in CxG, so, e.g., these two patterns are licensed by distinct constructions:
 - *There stood an old cabin.* (Presentational-*there*)
 - *An old cabin stood there.* (Subject-Predicate)

Goal: to explore how (and what) constructions mean

- Constructions and meaning composition
- Continuum of idiomaticity (with a focus on MWEs and illustration of the SBCG approach)
- Kinds of constructional meanings
- Argument structure (argument realization patterns)
- Conventional implicature
- Illocutionary force
- Metalinguistic commentary
- Focus articulation

Constructions and compositionality

- It is often said that CxG is not compositional, or ‘rejects semantic composition’.
- This is probably because CxG proponents focus on cases in which syntactic patterns mean what they mean in a word-like way: via convention.
- Such cases sometimes involve ‘constructional polysemy’ (non-syntactic ambiguity):
 - *She joked her way into the shareholders’ meeting* (manner or means reading; Jackendoff 1990, Goldberg 1995)
- But a compositional approach and a construction-based approach to syntax are actually compatible.

Constructions and compositionality

- In CxG, rules of syntactic combination (descriptions of local trees) are directly associated with interpretive/use conditions.
- These conditions are captured by semantic and pragmatic features attached to the mother or daughter nodes in these descriptions (Kay 2002; Sag 2010, 2012).
- This amounts to the claim that syntactic rules mean things.

Example: the *What* Exclamative cxn

Wh-Exclamative Construction ($\uparrow$ *filler-head-cxt* & $\uparrow$ *exclamative-cl*):

$$wh-excl-cl \Rightarrow \left[\begin{array}{l} \text{MTR} \quad [\text{SEM} \quad fact(\mathbf{Q}[\lambda \mathbf{X}[\mathbf{Y}](\mathbf{Z}))] \\ \text{DTRS} \quad \left\langle \begin{array}{l} \text{SYN} \quad \left[\begin{array}{l} \text{CAT} \quad nonverbal \\ \text{WH} \quad \{\mathbf{Q}\} \end{array} \right] \\ \text{SEM} \quad \mathbf{Z} \end{array} \right\rangle, \left[\begin{array}{l} \text{SYN} \quad \left[\begin{array}{l} \text{CAT} \quad \left[\begin{array}{l} \text{INV} \quad - \\ \text{VFORM} \quad fin \end{array} \right] \\ \text{VAL} \quad \langle \rangle \\ \text{GAP} \quad \langle [\text{SEM} \quad \mathbf{X}] \rangle \end{array} \right] \\ \text{SEM} \quad \mathbf{Y} \end{array} \right] \right\rangle \end{array} \right]$$

Constructions and compositionality

- Meaning is generally seen as something only words do.
- In the prevailing view, syntactic rules (PS rules) do only one thing: determine what symbol sequences function as units for syntactic purposes.
 - Syntactic rules assemble words and their dependent elements into phrases to be combined recursively.
 - The phrases denote complex concepts like predicates and propositions.
 - But the rules cannot add conceptual content to that contributed by the words (Jackendoff 1997).
 - And the rules cannot alter the combinatoric properties of the words.
- Jackendoff (1997) calls this view *syntactically transparent composition* (STC).

Constructions and compositionality

- According to STC, “[a]ll elements of content in the meaning of a sentence are found in the lexical conceptual structures [...] of the lexical items composing the sentence” (Jackendoff 1997: 49).
- CxG does not reject the existence of STC.
- Instead it treats STC as a “default in a wider array of options” (ibid).
- Whenever a class of expressions can be licensed by a context-free PS rule accompanied by a rule composing the semantics of the mother from the semantics of the daughters, CxG proposes a construction that is functionally equivalent to such a rule-to-rule pair.

Constructions and compositionality

- But CxG also offers a revealing way to represent linguistic structures in which the semantics of the mother does not follow entirely from the semantics of the daughters.
- One of the major motivations for CxG, as we have seen, is that a given rule of syntactic formation can sometime be associated with more than one semantic specification.
- A case in point is the Pseudo-conditional.

Pseudo-conditionals

- ***“If you’re Obama, you want your fingerprints nowhere near this thing,”** said Charles Freeman, Freeman Chair in China Studies at the Center for Strategic and International Studies. “It can only go badly”.*
- *If you’re 3Com right now, you’re considering buying ad space in next week’s issue.*
- *If you’re George Bush, you’re now allowed to lie in the faces of trusting young voters.*
- *If you’re Hillary Clinton and you get it into your head to publish an editorial, you can have access to pretty much any newspaper or magazine in the country...*

Pseudo-conditionals

- PCs have the same form as ordinary conditionals, e.g., *If you're rich, you probably think you deserve it.*
 - If you are x , $p(x)$
- But no hypothetical situation is posed.
- Instead, a categorical judgment is expressed; the subject of that judgment is not the addressee but the person identified as x .
 - *If you're Obama, you want your fingerprints nowhere near this thing.*
 - This example is about Obama, not about the consequences of a hypothetical identity between Obama and the addressee.
- PCs have the same form as (one type of) vanilla conditional but entirely distinct semantics.

Pseudo-conditionals

- If the grammar gives a sentence a different interpretation from what could be built up piece by piece from its words and constituent phrases, this is not strict syntactic composition.
- So the PC pattern could be called an idiom.
- But idiomaticity is not the same thing as inflexibility.
- The PC pattern is evidently a productive one.
- An adequate grammar must describe the interpretive and combinatoric constraints that define the PC, and other ‘funny patterns’.
- Doing so requires many more rules of composition than a non-constructional approach would ever allow—roughly as many as there are constructions listed in an (ideal) traditional grammar.

Pseudo-conditionals

- The PC is important to us because its existence undermines a foundational assumption of STC:
 - “If a language is compositional, it cannot contain a pair of non-synonymous complex expressions with identical structure and pairwise synonymous constituents.” (Szabó 2007)
- If we use Szabó’s diagnostic, the existence of the PC entails either:
 - English is not compositional; or
 - PCs are syntactically distinct from ordinary present-tense conditionals.
- And such a narrow view of compositionality also seems to require different syntactic analyses for any pair of readings of a sentence:
 - *My yoga instructor sometimes pulls my leg.*

Constructions and compositionality

- A constructional approach welcomes a single syntactic analysis in all such cases.
- It posits constructions in the case of the idiomatic readings that attach semantic interpretations directly to certain relatively complex syntactic objects.
- CxG also recognizes as instances of compositionality cases where two different meanings for the same syntactic form are licensed by two different collections of form-meaning licensers, i.e., by two different collections of constructions.

Constructions and compositionality

- ‘My yoga instructor sometimes pulls my leg.’

My yoga instructor sometimes [X]

Subject-Predicate Construction

pulls my leg

Head-Complement construction

pulls my leg

Idiomatic *pull* listeme
Head-Complement construction

Constructions and compositionality

- Constructional approaches tend to focus on the fact that there are many meaning-assigning rules, and especially on the rules that assign meanings to complex structures.
- In this approach, we do not draw either of these theoretical distinctions:
 - Between rules thought to be ‘in the core’ and those considered ‘peripheral’.
 - Between rules that assign ‘idiomatic’ interpretations and those that don’t.

The continuum of idiomaticity

- The construction grammarian conceives of a language as presenting a continuum of idiomaticity, or generality, of expressions.
- A construction grammar models this continuum with an array of constructions of correspondingly graded generality.

Continuum of idiomaticity overview

The salt of the earth
In the doghouse
Under the weather

Subject-Predicate
Construction
Kim blinked.



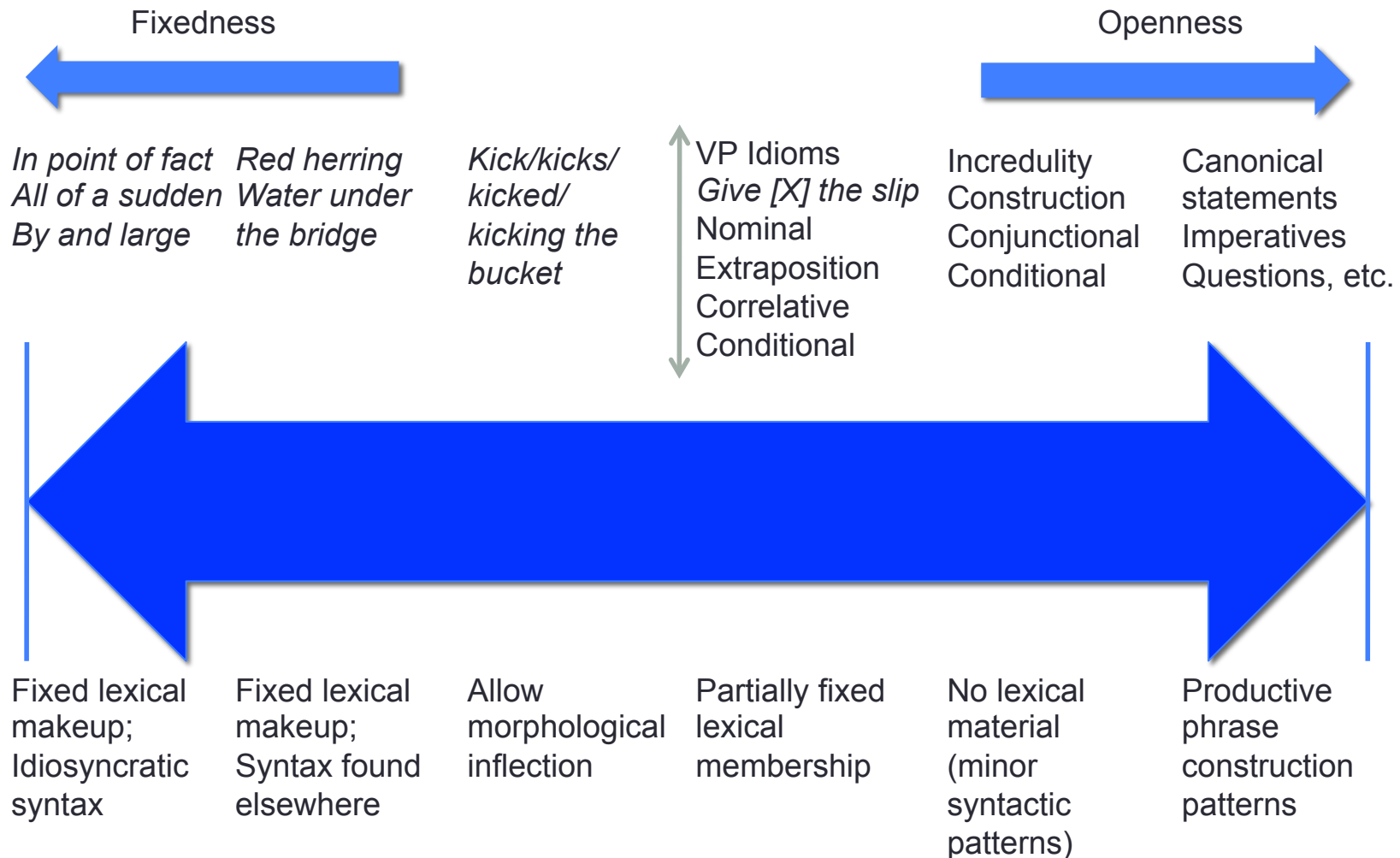
Frozen Idioms

Fully Productive
Rules

Idioms

- Defined (per Makkai 1972) as:
 - Expressions that are not (readily) interpretable by a naïve speaker (decoding idioms): *The more, the merrier. Not in my house you don't. Now watch it rain. Atta boy.*
 - Expressions that a naïve speaker would not know to use in a given context (encoding idioms): *bright red, deep voice, heavy smoker, answer the door, perform surgery.*

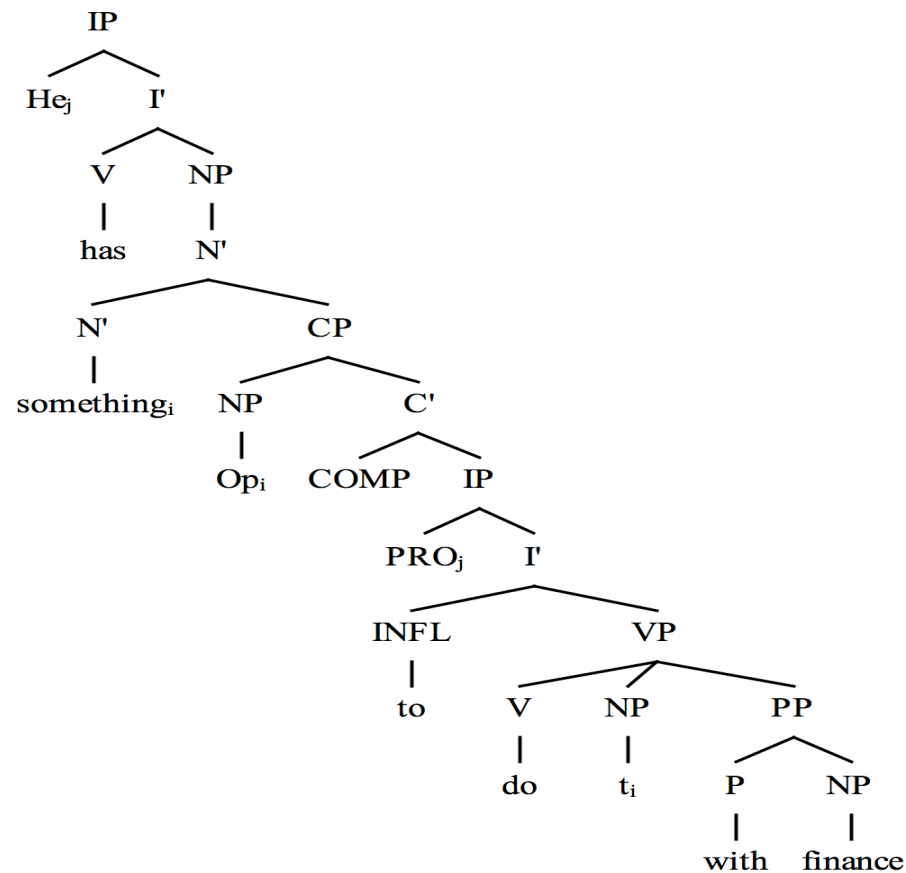
Continuum of idiomaticity



Locality and idioms: what's love got to do with it?

- Many idioms, including those we would intuitively regard as MWEs, appear to violate locality of selection.
- Locality says that a lexical head can select its sisters but not daughters of sisters.
- One apparently non-local MWE is *have x to do with y*
 - *Nobody wanted to **have anything to do with** this dog because it reeked of fish and everything else.*
 - *Well, yeah nine eleven and and saddam hussein **have nothing to do with** each other.*
 - *Most of the criticism of the book **has to do with** Ana's vanilla, generic and cheesy response to Christian's world.*
 - *What does this **have to do** with my life?*

Locality and hxtldwy

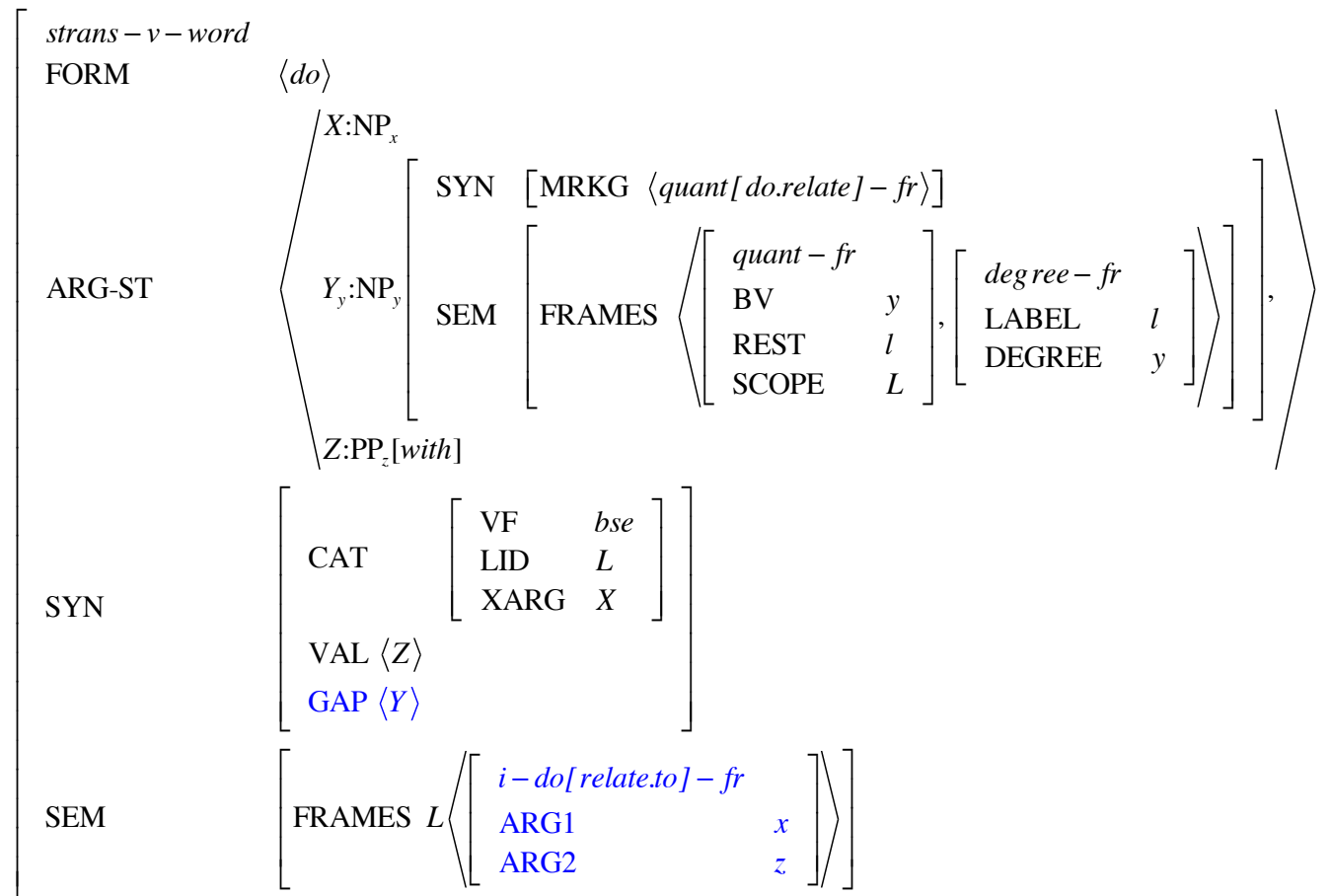


The verb *have* takes an object argument consisting of an indefinite word plus an infinitival relative clause whose main verb must be *do*. This *do* must take a PP headed by *with*.

Locality and hxtldwy

- The locality issue disappears if we ‘shave down’ the idiom.
- We can do this because the quantified nominal head (e.g., *little*, *a lot*, *more*, *nothing*, *something*) can be the complement of other verbs besides *have*:
 - **Found** *something to do with Astrology* that is actually quite good.
 - In 6th-grade, no one **wanted** *anything to do with me*.
 - *It is* *nothing to do with your attitude*.
- So let’s suppose the construction consists of the **quantified nominal modified by an infinitival relative clause containing *do* and the PP *with***.
- Then the idiom boils down to the direct object and PP complement selected by (idiomatic) *do*.
- This is the kind of ‘lexically headed’ analysis of idioms proposed by Kay and Sag (forthcoming).

Idiomatic *do* listeme



Kinds of constructional meanings

- Truth-conditional meaning (won't discuss this)
- Event types as expressed by constellations of arguments
- Conventional implicatures
- Illocutionary forces, including less commonly recognized ones (LCRIFs)
- Metalinguistic commentary
- Focus articulation (discourse-pragmatic content)

Form-meaning pairings

- A construction is never pure form nor pure meaning, but a pairing of the two (Zwicky 1994).
- Meanings are attached to various kinds of constructions (and listemes).
 - **Combinatoric constructions:** these describe constructs (mother-daughter combinations) like Subject Predicate, Head Complement, etc.
 - **Derivational constructions:** these allow new lexemes to be built from one or more lexical signs, e.g., nominal compounds.
 - **Lexical-class constructions:** these describe lexical classes with characteristic combinatoric behaviors, e.g., the class of ditransitive lexemes.
 - **Listemes:** a description of a lexical sign or **idiomatically combining expression**.

Two kinds of idiomatically combining expressions

- Semantically decomposable idioms are MWEs whose idiomatic meaning is distributed over the meaning of the sub-expressions: *spill the beans*, *keep tabs on*, *pull strings* (Nunberg et al. 1994, Sag et al. 2002).
- Only decomposable idioms are syntactically flexible:
 - *Many **strings** were pulled.*
 - **Many **buckets** were kicked.*

$$\left[\begin{array}{l} \text{cn-lxm} \\ \text{FORM} \quad \langle \text{strings} \rangle \\ \text{SEM} \quad \left[\begin{array}{l} \text{IND} \quad i \\ \text{FRAMES} \quad \left\langle \left[\begin{array}{l} i\text{-strings}^{\text{connections}}\text{-fr} \\ \text{ENTITY} \quad i \end{array} \right] \right\rangle \end{array} \right] \end{array} \right]$$

$$\left[\begin{array}{l} \text{cn-lxm} \\ \text{FORM} \quad \langle \text{bucket} \rangle \\ \text{SYN} \quad [\text{CAT} [\text{LID} \langle i\text{-bucket-fr} \rangle]] \\ \text{SEM} \quad \left[\begin{array}{l} \text{IND} \quad \text{none} \\ \text{FRAMES} \quad \langle \rangle \end{array} \right] \end{array} \right]$$

Signs

- Form and meaning are of course the two poles of a sign.
- In SBCG, language is an infinite set of signs (modeled as feature structures).
- Signs are analyzed as FSs that specify values for PHON, FORM, SYN, SEM, and CNTXT.
- So the grammar signature contains this type declaration:

$$\text{sign} : \begin{bmatrix} \text{PHON} & \textit{phon-obj} \\ \text{FORM} & \textit{morph-obj} \\ \text{SYN} & \textit{syn-obj} \\ \text{SEM} & \textit{sem-obj} \\ \text{CNTXT} & \textit{context-obj} \end{bmatrix}$$

The Sign Principle

- The array of signs includes words, lexemes and phrases.
- Signs are constrained by the Sign Principle (Sag 2012):
 - Every sign must be listemically or constructionally licensed, where:
 - a sign is listemically licensed only if it satisfies some listeme, and
 - a sign is constructionally licensed only if it is the mother of some well-formed construct.

Kinds of constructional meanings

- Event types as expressed by constellations of arguments
- Conventional implicatures
- Less commonly recognized illocutionary forces (LCRIFs)
- Metalinguistic commentary
- Focus articulation (discourse-pragmatic content)

Argument structures

- Argument structure constructions (ASCs) regulate the expression of a verb's arguments.
- SBCG models ASCs as both lexical-class and derivational constructions.
- ASCs denote coarse-grained event types, and thus feature their own argument arrays.
- Accordingly, they can **alter** the array of arguments that the verb selects for intrinsically:
 - The Caused-Motion construction
 - *The kids **swam** the logs upstream.*
 - The Locative Inversion construction
 - *From the mast **flapped** the banner of King Aurelius and of Braime.*
 - The Ditransitive construction
 - Santa brought a baby Lab and last week we **adopted** her a sister at Devore shelter.

Conventional implicatures

- In CxG, a Gricean conventional implicature is seen as a directive to the addressee: extract information of a prescribed type from the context.
- Kay (1990) refers to words and constructions that carry conventional implicatures as *contextual operators*.
- Their “semantic value consists [...] of instructions to find in the context a certain kind of information structure and place the information presented by the sentence within that information structure” (Kay 1997:159):
 - *The DCCC Won’t Acknowledge Him, Let Alone Help Him, But There Is A Democrat Running Against John Boehner This Year.*
 - Scalar model: *let alone* “contextual operator” instructs addressee to find in context a scalar model in which the first clause (TP) entails the second, CP (Fillmore et al. 1988)
- Such cases suggest that conventional implicatures are part of meaning assembly: we can point to the syntactic component that carries the instruction, e.g., the conjunction *let alone*.

Contextual operators are not just words

- Kay (1997) notes a *why*-conditional sentence type that evokes a *modus tollens* inference pattern:
 - If you're so smart, why aren't you rich?
- By using the *why*-conditional, the speaker rejects the truth of the proposition in the antecedent (you're smart) by inducing the hearer to accept:
 - The truth of the consequent (you aren't rich)
 - A causal stereotype (smart people get rich).
- This rhetorical strategy has become a conventional function of this conditional form: ask the hearer to perform the *modus tollens* syllogism in order to cast doubt on the truth of a proposition in the common ground.
- The use of *so* in the antecedent distinguishes conventional *modus tollens* from an innocent *why* question (one that preserves commitment to addressee's smartness).

Less commonly recognized illocutionary forces (IFs)

- In SBCG, contextual restrictions on constructions are represented by the CONTEXT feature, whose values include BACKGROUND.
- Values of BACKGROUND correspond to utterance felicity conditions.
- Utterance felicity conditions can define a wide array of illocutionary forces (publically stated commitments):
 - Familiar IF: imperatives, questions
 - Special IF: expression of incredulity
 - *Him get first prize?!*
 - Special IF: 'conjuring fate'
 - *Watch me (get wet/fall/drop it)*
 - Special IF: desire for explanation of anomaly
 - *What's this fly doing in my soup?*
 - Special IF: ??
 - *If you're Britney Spears' publicist, you might as well go ahead and kill yourself*

Metalinguistic comments

- Metalinguistic constructions comment on the form or meaning (or both) of the utterance in which they occur.
- Example: metalinguistic negation
 - *It's not the unique criteria; it's the unique criterion*
 - *I'm not sweating, I'm perspiring.*
- None express negation of a proposition.
- Rather such utterances object to an aspect of the utterance *other* than the propositional content
- Special syntactic form is evidence of a construction: negated clause (that doesn't negate truth of proposition), followed by rectification clause.

Metalinguistic comments

- Another example: **hedges** (Lakoff 1972, Kay 1982)
- *Chomsky has a very sorta classical theory of syntax.*
- Adverb *very* intensifies adjective *classical*, but hedge *sorta* is metalinguistic: signals that speaker is unsure of choice of term *classical*
- Hedges are syntactically distinguishable from ordinary de-intensifiers like *slightly*:
 - *a slightly but unevenly worn tire*
 - **a sorta but unevenly worn tire*
 - *That tire is worn slightly.*
 - **That tire is worn sorta*
- Special syntactic and semantic properties make it appropriate to speak of metalinguistic constructions.

Focus articulations

- CxG approach recognizes three basic focus constructions, with prosodic manifestations in English:
 1. **Argument focus:** *SOCIETY'S to blame* (identifies a variable in a presupposed open proposition)
 2. **Predicate focus:** *She speaks several LANGUAGES* (predicates a property of a topical entity)
 3. **Sentence focus:** *Your SHOE'S untied* (introduces a new discourse referent, reports an event or state involving that referent)
- Prosodic permutations amount to differences in presuppositional content of the constructions that license them.
 - Knowledge presuppositions
 - Topicality presuppositions
- Predicate focus construction triggers the topicality presupposition.
- Users exploit this topicality presupposition in order to pack two discourse moves into one clause.

Focus articulations

- A speaker can use the PF construction when the topicality presupposition is not satisfied:
 - [In a conversation about vacations] *Uh right exactly so this summer **um my boyfriend lives in California** and he loves to go camping...*
- This is a form of presupposition exploitation: acting as if a proposition ('I have a boyfriend') is already part of the common ground (Stalnaker 1974).
- This referent-introduction strategy is more economical (although rarer in conversation) than the two-clause strategy:
 - *Or, you know, I have **a friend of mine** and uh he hasn't seen one of his cats for, you know, like going on six weeks now.*
- When a pragmatic constraint is attached to a construction, it can be exploited for pragmatic purposes.

Conclusions

- In asking what constructions mean we must also ask how constructions mean.
- Constructions invoke formal properties ranging from syntactic categories to prosodic features to fixed lexical forms.
- All such patterns must interact in the licensing of utterances.
- The recursive nature of a language comes from the fact that we can use in one construction a sign that is an instance of another construction.

Conclusions

- No current syntactic theory fails to acknowledge that verbal idioms and their ilk can be embedded as the terminal nodes of regularly constructed phrases.
- But non-constructionists have been less apt to acknowledge another fact about embedding: regular patterns can be embedded in idiomatic ones.

Conclusions

- Examples include:
 - *Like*-prefixed sarcasm (Camp 2012): *Like I care what you do, As if YOU could fix it.*
 - This construction is idiomatic: subordinate clauses aren't supposed to serve main-clause functions, but the portion following the conjunction (*like, as if*) is a regular subject-predicate clause.
 - Subjectless tag sentences (Kay 2002): *Scared you, didn't I?*
 - This construction is idiomatic: a finite clause is supposed to have as subject, but the tag is a regular polar tag.
 - The double-copula construction (Brenier & Michaelis 2005): *The thing is is that the Iranians aren't helping the situation.*
 - This construction is idiomatic: we don't otherwise find two finite forms of a verb in a row, without any subordination, but the VP following BE1 is an ordinary VP.

Conclusions

- The seamless integration of relatively idiomatic constructions with more productive ones in actual sentences provides an additional challenge to the notion of a privileged 'core' grammar.
- There is no fire wall separating idiomatic constructions from regular constructions.



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